Symmetries in Physics — Exercise Sheet 15

The Poincaré group and particles

Variant: Questions only

Exercise 1 Derive three representative Poincaré brackets by hand from the 5×5 matrix model, in which a Poincaré element is $\left(\begin{smallmatrix} \Lambda & a \\ 0 & 1 \end{smallmatrix}\right)$ acting on $\left(\begin{smallmatrix} x \\ 1 \end{smallmatrix}\right)$. Compute $[p_1, p_2]$, $[J_3, p_1]$, and $[K_3, p_0]$ for the abstract Lie-algebra basis, then translate them to the self-adjoint observables of Lecture 15.

Exercise 2 On a common invariant domain for the derived operators (take the smooth compactly supported V_j-valued functions on the mass shell, which all the generators of Lecture 15's induced representation preserve), start from the observable Poincaré brackets and the symmetrised Pauli–Lubański vector. Prove explicitly that $[\widehat{W}^\mu, \widehat{P}_\nu] = 0$ and compute the full bracket $[\widehat{M}_{\alpha\beta}, \widehat{W}^\mu]$. Use the product rule to prove that $\widehat{W}^2 = \widehat{W}_\mu \widehat{W}^\mu$ commutes with every Poincaré generator. Finally compute its value for a massive spin-j particle in the rest frame. You may use the epsilon-invariance identity $\Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta \Lambda^\rho{}_\gamma \Lambda^\sigma{}_\delta \, \epsilon^{\alpha\beta\gamma\delta} = (\det \Lambda)\, \epsilon^{\mu\nu\rho\sigma}$ in its infinitesimal, indexed form, which you should derive, but may not merely assert covariance.

Exercise 3 Find the little group of a massless momentum. Represent $p = (E, 0, 0, E)$ by the Hermitian matrix $X(p) = E(\not{\mathbb{1}} + \sigma_3) = \mathrm{diag}(2E, 0)$, find the $A \in \mathrm{SL}(2, \mathbb{C})$ fixing it, and identify the resulting three-parameter group as $\widetilde{\mathrm{E}}(2) = \mathbb{C} \rtimes \mathrm{U}(1)$, the double cover of the geometric Euclidean plane group, with $\alpha : \beta \mapsto \alpha^2 \beta$. Then prove that translations act trivially in every finite-dimensional unitary representation. State, using Lecture 15's separately imported little-group classification, what happens to nonzero translation-character orbits.

Exercise 4 Show the Lorentz-invariance of the on-shell measure $\dfrac{\mathrm{d}^3 p}{2E_p}$ under a boost along z, first for the massive shell $m > 0$ and then for the punctured forward light cone $m = 0$, $E_p = |\vec{p}| > 0$. Explain why the cone vertex is excluded.

Exercise 5 Verify the Wigner-rotation cocycle in a two-boost example, closing the loop with Sheet 14: show that boosting a rest particle by A_2 then A_1 produces a little-group element equal to the Thomas–Wigner rotation.

Exercise 6 For a massless particle, show that $R_z(\theta)$ acts on a helicity-h state by the phase $e^{-ih\theta}$, and explain why $h \in \frac{1}{2}\mathbb{Z}$. Determine the phase of the central 2π rotation. Do not infer exchange statistics from that phase.